

Operating Systems Principles: Memory Management & Paging

Section 1: Address Translation Mechanism

Modern CPU architectures rely on hardware paging units to map virtual addresses to physical addresses.

The page table maintains mapping entries containing physical frame numbers and permission bits.

Translation Lookaside Buffers (TLBs) cache recent translations to minimize page table walk latencies.

Section 2: Page Replacement Algorithms

When a page fault occurs and no physical frames are free, the kernel must evict an existing page.

Common replacement policies include Least Recently Used (LRU), Clock Algorithm, and Second-Chance.

Dirty pages must be written back to backing store before frame reclamation.